

CloudShift - Software Management | Risk Register and Closure Note

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1. Probability-Impact Risk Matrix (5x5 Grid)

The matrix below maps project risks across a 5x5 Probability (P, scale 1–5) vs. Impact (I, scale 1–5) grid. Risk Exposure score is calculated as $E = P \times I$.

- **High Exposure (E = 15–25):** Requires immediate proactive RMMM mitigation plan.
- **Medium Exposure (E = 8–12):** Requires active monitoring and contingency triggers.
- **Low Exposure (E = 1–6):** Acceptable risk; monitored via standard logs.

Probability / Impact	1: Negligible	2: Minor	3: Moderate	4: Major	5: Critical
5: Very High	Score: 5 (LOW)	Score: 10 (MED)	Score: 15 (HIGH)	Score: 20 (HIGH)	Score: 25 (HIGH)
4: High	Score: 4 (LOW)	Score: 8 (MED)	Score: 12 (MED)	Score: 16 (HIGH)	[R-01] Score: 20 (HIGH)
3: Medium	Score: 3 (LOW)	Score: 6 (LOW)	Score: 9 (MED)	[R-04] Score: 12 (MED)	[R-02] Score: 15 (HIGH)
2: Low	Score: 2 (LOW)	[R-06] Score: 4 (LOW)	[R-05] Score: 6 (LOW)	Score: 8 (MED)	[R-03] Score: 10 (MED)
1: Very Low	Score: 1 (LOW)	Score: 2 (LOW)	Score: 3 (LOW)	Score: 4 (LOW)	Score: 5 (LOW)

2. Risk Register Table

Exposure Score ($E = P \times I$) where Probability (P, scale 1–5) and Impact (I, scale 1–5). The three highest-exposure risks in the supplied register are R-01 ($E = 20$), R-02 ($E = 15$), and R-04 ($E = 12$).

Risk ID	Risk Description	Category	P	I	Exposure (E)	Response Strategy	Risk Owner
R-01	Undocumented Legacy Code (App7-9): Hidden hardcoded dependencies or missing source files delay migration.	Technical Debt	4	5	20 (HIGH)	Mitigate (Reverse eng, PERT estimation, containerization)	Eng 3 / Lead
R-02	Blackout Window Violation: Migration or cutover bleeds into last 5 working days of month, interrupting billing.	Operational	3	5	15 (HIGH)	Avoid (Automated Blackout Guard lock service)	Eng 2 / Infra
R-04	On-Premise Server Hardware Failure: Ageing servers crash before 7-month contract expiry.	Infrastructure	3	4	12 (MEDIUM)	Mitigate (Prioritize high-risk apps in early waves)	Cloud Eng
R-03	Data Loss during Cutover: Database replication desynchronization causes missing transactions.	Data Integrity	2	5	10 (MEDIUM)	Mitigate (CDC real-time sync, SHA-256 pre-cutover hash audit)	Eng 1 / DBA
R-05	DNS Propagation Lag: Extended DNS cached TTL causes temporary access drop for end users.	Networking	2	3	6 (LOW)	Mitigate (Reduce Route53 TTL to 60s 48h prior to cutover)	Network Lead
R-06	Cloud Cost Overrun: Provisioned cloud resources exceed initial monthly operating budget.	Financial	2	2	4 (LOW)	Accept (Set up CloudWatch budget alerts and auto-scaling)	CIO

3. RMMM Plans for Top 3 Ranked Risks (R-01, R-02, R-04)

3.1 Risk 1 (R-01): Undocumented Legacy Codebase (App7, App8, App9)

- **Risk Exposure:** $E = 4 \times 5 = 20$ (Rank #1)
- **Risk Mitigation:**
 - Conduct static code analysis and dynamic memory profiling to discover runtime dependencies.
 - Encapsulate legacy runtimes in Docker container images to bypass OS-level library incompatibilities.
 - Apply 3-point PERT estimation (TE = 22 person-days) with a 95% confidence buffer.
- **Risk Monitoring:** Track daily execution velocity against PERT estimates. If reverse engineering exceeds 15 person-days per app, trigger secondary senior developer review.
- **Risk Management / Contingency:** Fall back to re-hosting (lift-and-shift VM encapsulation) if code refactoring stalls beyond threshold.

3.2 Risk 2 (R-02): Month-End Blackout Window Violation

- **Risk Exposure:** $E = 3 \times 5 = 15$ (Rank #2)
- **Risk Mitigation:**
 - Program BlackoutGuard microservice directly into the CI/CD deployment pipeline.
 - Implement automated pipeline locking triggered on Day 18 of every working month.
- **Risk Monitoring:** Real-time countdown timer displayed on the CloudShift Executive Dashboard (FR-12).
- **Risk Management / Contingency:** If cutover is incomplete by 23:59 on Day 17, automatically abort deployment and defer cutover to Day 1 of the following month.

3.3 Risk 3 (R-04): On-Premise Server Hardware Failure

- **Risk Exposure:** $E = 3 \times 4 = 12$ (Rank #3)
 - **Risk Mitigation:**
 - Prioritize migration sequence for mission-critical apps on oldest hardware servers in Wave 1 and Wave 2.
 - Maintain pre-configured IaC cloud landing zone ready for emergency instant workload deployment.
 - Perform daily full database and image backups of all remaining on-premise hosts.
 - **Risk Monitoring:** Daily automated SMART health monitoring of disk arrays and CPU temperature logs on legacy office servers.
- Risk Management / Contingency:** If on-premise hardware experiences critical failure prior to scheduled wave cutover, trigger emergency accelerated lift-and-shift migration to cloud staging environment.

4. Separate Historical Issue Log (Past Occurrences)

This log isolates problems that **have already occurred** in legacy on-premise operations prior to project CloudShift.

Issue ID	Incident Description	Occurrence Date	Impact / Outage	Root Cause Analysis	Preventative Guardrail	Status
ISS-01	Billing Application Server Failure	Prior Year	2-Day Total Outage of billing app	Unhandled hardware RAID controller failure on ageing office server without failover.	Replaced single server with multi-AZ cloud auto-scaling group (NFR-01: 99.95% availability).	RESOLVED
ISS-02	On-Premise Power Supply Degradation	6 Months Ago	4-Hour Intermittent Outage	UPS battery failure during office building power fluctuation.	Cloud migration eliminates local power dependency; cloud multi-region backup implemented.	RESOLVED

5. Project Closure Note & Lessons-Learned Summary

Note: The following closure results represent the simulated completion of the CloudShift case study based on the defined requirements, estimates, and test evidence.

5.1 Key Lessons Learned

2. **Automated Policy Enforcement is Essential:** Relying on manual developer promises to avoid month-end blackouts failed in past projects. Hardcoded programmatic pipeline locks (BlackoutGuard) eliminate human error.
3. **PERT Estimation Mitigates Technical Debt:** Undocumented legacy applications (App7–9) carry fat-tailed risk distributions ($P = 40$ days). PERT three-point calculations successfully provided realistic schedule buffers ($TE = 22$ days).
4. **Automated Rollback Prevents Outages:** Designing an automated $RTO \leq 15$ min rollback mechanism is designed to limit the duration of a failed cutover and reduce the chance of a prolonged outage.

5.2 Operational Handoff Checklist

- 100% of 9 applications live in public cloud infrastructure.
- All 8 NFR targets verified via synthetic tests and load benchmarks.
- On-premise office hardware powered down and prepped for disposal.

Support contracts formally transferred to Cloud Operations Team.

6. Refinement Notes

The risk register keeps **risks** separate from the historical **issue log**. Risks describe possible future events; ISS-01 and ISS-02 record incidents that have already occurred.

The probability-impact exposure score is strictly:

Exposure = Probability × Impact

The RMMM actions for the three highest-exposure risks are retained because the deliverable explicitly asks for RMMM treatment of the top three exposures. The closure checklist is labelled as a **simulated case-study completion state**, consistent with the source document.

Probability-Impact Matrix

